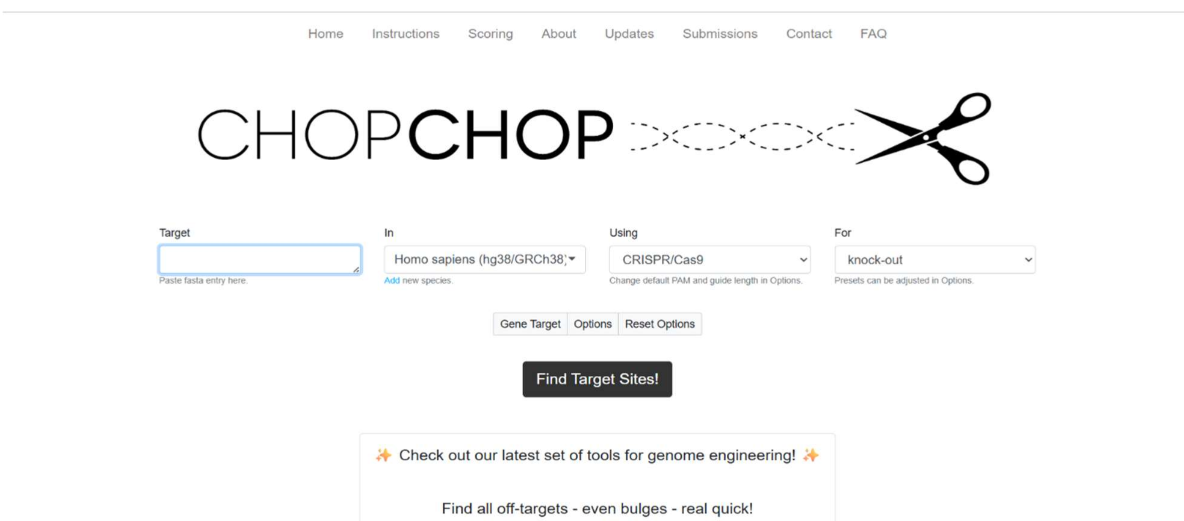
Figures determining Steps & Result

1. **Get your gene:**

   Choose your target gene i.e. PCSK9

2. **Visualization using SnapGene Viewer:**

3. **Designing Guides using CHOPCHOP:**



4. Run & Review possible guide sequences:

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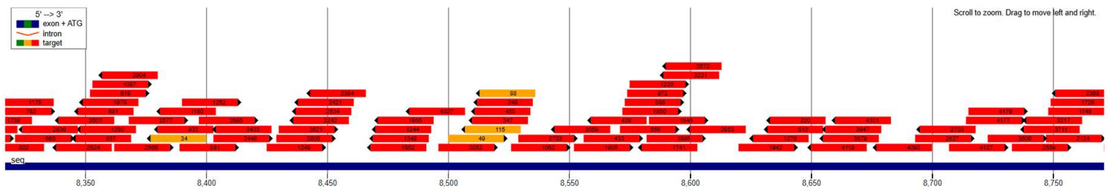
fastaInput.fa



Download results: Please select one

View in UCSC genome browser

Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	GCCCCATGTCGACTACATCGAGG	seq:6987	+	60	1	1	0	0	0	75.32
2	TTAATCAGCTCAGCATACCGTGG	seq:5294	+	45	0	1	0	0	0	69.63
3	AAGGGGCGGAGGTATTCTCGAGG	seq:1131	+	60	1	1	0	0	0	68.66
4	TACCCCTCCACGGTACCGGGCGG	seq:7059	+	70	2	1	0	0	0	66.34
5	GACCTACTAGCATAGTGGGTGGG	seq:11132	+	50	2	1	0	0	0	64.58
6	GAGCAAACTCGCCCCGCACACGG	seq:1280	-	65	0	1	0	0	0	58.90
7	CTCCTCGATGTAGTCGACATGGG	seq:6989	-	50	2	1	0	0	0	60.82
8	AAACTCGCCCCGCACACGGTCGG	seq:1276	-	65	0	1	0	0	0	58.82
9	ATCCCACGAGAACCGGTAAGGGG	seq:7899	-	55	0	1	0	0	0	58.36

5. Scoring & Evaluating the Guides in Excel using required filters:

Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	GCCCCATGTCGACTACATCGAGG	seq:6987	+	60	1	1	0	0	0	75.32
2	TTAATCAGCTCAGCATACCGTGG	seq:5294	+	45	0	1	0	0	0	69.63
3	AAGGGGCGGAGGTATTCTCGAGG	seq:1131	+	60	1	1	0	0	0	68.66
9	ATCCCACGAGAACCGGTAAGGGG	seq:7899	-	55	0	1	0	0	0	58.36

6. Copy the Guides from Excel and view it in SnapGene Viewer.

- *Guide_RNA.Cas9*

➢ *Guide_RNA.Cas12a*

➢ *Overall Result View*